

Text Summarization across different platforms

Name: Harish Kumar P

Register no:212225230095

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Subject:19TD608- PROMPT ENGINEERING

Text Selected for Summarization:

Blockchain technology is a decentralized, distributed, and immutable digital ledger designed to record transactions across a network of computers, ensuring data integrity without a central authority. At its core, it functions as a chronological chain of data blocks, where each block contains a batch of transactions, a timestamp, and a unique cryptographic hash of the preceding block. This cryptographic linkage makes the ledger tamper-evident; altering any historical data would require recalculating all subsequent blocks, which is computationally infeasible.

The network operates on a peer-to-peer (P2P) architecture, where participants, known as nodes, maintain a synchronized copy of the entire ledger. To ensure data consistency across these distributed nodes, the system utilizes consensus mechanisms—such as Proof of Work (PoW) or Proof of Stake (PoS)—to validate new transactions and reach agreement on the state of the network. This decentralized validation removes the need for trusted intermediaries, effectively eliminating single points of failure and reducing risks associated with double-spending.

Beyond basic transaction storage, blockchain technology often incorporates smart contracts, which are self-executing scripts that automate actions when predefined conditions are met. This combination of immutability, decentralization, and transparency provides a secure infrastructure for diverse applications, ranging from cryptocurrency to supply chain management and decentralized finance.

AI Platforms Used:

- ChatGPT (OpenAI)
- Gemini (Google)
- Copilot (Microsoft)

Execution:

Each platform received the same article and was prompted with each technique. Outputs and processing times were recorded for comparison.

Evaluation Criteria:

- **Accuracy:** Correctness in capturing the article's key points.
- **Coherence:** Logical sequencing and readability.
- **Simplicity:** Accessibility of language for undergraduate students.
- **Speed:** Time taken to generate the summary.
- **User Experience (UX):** Interface intuitiveness and ease of saving output.

Each criterion was scored on a scale from 1 (poor) to 5 (excellent).

Prompting Techniques Defined:

1. Zero-Shot Prompting:

Prompt:

- Summarize the text without giving any examples.

Responses generated by AI Tools:

⇒ ChatGPT:

Blockchain technology is a decentralized and distributed digital ledger that records transactions across a network of computers without relying on a central authority. It organizes data into sequential blocks linked through cryptographic hashes, making the ledger secure and resistant to tampering. A peer-to-peer network of nodes maintains synchronized copies of the ledger and uses consensus mechanisms to validate transactions and maintain consistency. The technology also supports smart contracts, enabling automated actions based on predefined conditions while providing a secure and transparent system for digital record-keeping.

⇒ Gemini:

Here is a summary of the key concepts from the text:

Blockchain Technology Overview

- **Core Concept:** It is a decentralized, immutable digital ledger that securely records transactions across a peer-to-peer network without relying on a central authority.
- **Data Security:** Information is structured in a chronological chain of cryptographically linked blocks. This design makes the system tamper-evident, as altering historical data is computationally unfeasible.
- **Network Operation:** Distributed nodes maintain a synchronized copy of the ledger by using consensus mechanisms to validate new transactions and agree on the network's state. This eliminates the need for trusted intermediaries and removes single points of failure.
- **Automation:** The technology utilizes smart contracts—self-executing scripts—to automatically perform actions when predefined conditions are met, creating a secure, transparent, and decentralized infrastructure.

⇒ **Copilot:**

Blockchain technology is a decentralized and immutable digital ledger that records transactions in a chronological chain of data blocks. Each block is cryptographically linked to the previous one, making alterations computationally impractical. The system operates on a peer-to-peer network where nodes maintain synchronized copies of the ledger, and consensus mechanisms ensure data consistency and validation. It also supports automated execution through smart contracts, combining transparency, security, and decentralization to provide a reliable infrastructure for digital interactions.

2. Few-shot Prompting:

Prompt:

- Summarize the text with giving two or three examples.

Responses generated by AI Tools:

⇒ ChatGPT:

Blockchain technology is a decentralized and distributed digital ledger that records transactions across a network without a central authority. It stores data in linked blocks secured by cryptographic hashes, making the system transparent and tamper-resistant. Transactions are verified through consensus mechanisms and maintained by a network of nodes, ensuring reliability and security. It can also run smart contracts that automatically execute actions when certain conditions are met.

Examples of its applications include **cryptocurrency transactions**, **supply chain tracking**, and **decentralized finance systems**.

⇒ Gemini:

- **Blockchain Technology Summary**
- **Core Concept:** Blockchain is a decentralized, immutable digital ledger that securely records transactions across a peer-to-peer network, completely removing the need for a central authority.
- **Security and Consensus:** Data is structured in a chronological chain of cryptographically linked blocks. Distributed nodes maintain the ledger and use consensus mechanisms (such as Proof of Work or Proof of Stake) to validate new transactions, making the system tamper-evident and eliminating single points of failure.
- **Automation:** The network often utilizes smart contracts—self-executing scripts that automatically trigger actions when predefined conditions are met.
- **Examples of Blockchain Applications** As highlighted in the text, this combination of decentralization, immutability, and transparency provides a secure infrastructure for diverse uses, such as:
 - **Cryptocurrency**
 - **Supply chain management**

- **Decentralized finance**

⇒ **Copilot:**

Blockchain technology is a decentralized and immutable ledger that records transactions in linked blocks, secured by cryptographic hashes. It operates on a peer-to-peer network where nodes maintain synchronized copies, and consensus mechanisms like Proof of Work or Proof of Stake validate new entries. In addition to transaction recording, it supports smart contracts—self-executing scripts that automate actions. This combination of transparency, security, and decentralization enables applications such as cryptocurrency, supply chain management, and decentralized finance.

3. Chain Of Thought Prompt:

Prompt:

• **Based on the above two summarization, give me the detailed summary of block chain Technology.**

Responses generated by AI Tools:

⇒ **ChatGPT:**

Blockchain technology is a **decentralized and distributed digital ledger** that records transactions across a network of computers without the need for a central authority. Instead of relying on a single controlling organization, the system is maintained by multiple participants in a peer-to-peer network, ensuring transparency and reducing the risk of manipulation or failure. Each participant, known as a node, keeps a synchronized copy of the ledger so that the data remains consistent and accessible across the entire network.

The information in a blockchain is stored in **blocks**, which are groups of transactions. Each block contains a timestamp, transaction data, and a unique cryptographic hash that links it to the previous block. This linking

forms a continuous chain of blocks. Because every block is connected to the one before it, any attempt to alter earlier data would require changing all the following blocks in the chain, making the system highly secure and resistant to tampering.

To maintain agreement among the distributed nodes, blockchain networks use **consensus mechanisms** such as Proof of Work (PoW) or Proof of Stake (PoS). These mechanisms allow the network to verify and validate transactions before they are added to the ledger. As a result, the system can operate securely without relying on trusted intermediaries and helps prevent issues like fraudulent transactions or double-spending.

Another important feature of blockchain technology is the use of **smart contracts**. Smart contracts are self-executing programs stored on the blockchain that automatically perform actions when specific conditions are met. This automation improves efficiency, reduces the need for manual verification, and increases reliability in digital transactions.

Due to its decentralization, transparency, and strong security features, blockchain technology is widely used in many fields. Common applications include **cryptocurrency systems**, where digital currencies are transferred securely; **supply chain management**, where product movement and authenticity can be tracked; and **decentralized finance (DeFi)** platforms, which enable financial services without traditional intermediaries. Overall, blockchain provides a reliable and secure framework for storing and managing digital information in a transparent and distributed environment.

⇒ **Gemini:**

- Detailed Summary of Blockchain Technology
- 1. Core Architecture and Concept Blockchain is a decentralized, distributed, and immutable digital ledger. It operates on a peer-to-peer (P2P) network where participants, known as nodes, each maintain a synchronized copy of the entire ledger. This structure completely removes the need for a central authority or trusted intermediaries to manage transactions.

- 2. Data Structure and Security Information is securely stored in a chronological chain of data blocks. Each block is built with three main components:
 - A batch of transactions.
 - A timestamp.
 - A unique cryptographic hash linking it to the preceding block.
- This cryptographic linkage makes the blockchain tamper-evident. Altering any historical data is computationally unfeasible, as it would require maliciously recalculating the hashes for that block and all subsequent blocks.
- 3. Consensus and Validation To keep the distributed ledger perfectly synchronized, the network relies on consensus mechanisms, such as Proof of Work (PoW) or Proof of Stake (PoS). These protocols allow distributed nodes to validate new transactions and agree on the current state of the network. This decentralized validation process effectively eliminates single points of failure and mitigates risks like double-spending.
- 4. Automation via Smart Contracts Beyond simply storing transactions, blockchain technology often integrates "smart contracts." These are self-executing scripts programmed into the blockchain that automatically trigger specific actions once predefined conditions are met, adding a layer of secure, programmable automation to the network.
- 5. Practical Applications The combination of decentralization, immutability, transparency, and automated smart contracts creates a highly secure infrastructure utilized across various industries. Notable examples include:
 - Cryptocurrency: Enabling secure, peer-to-peer digital payments without banks.
 - Decentralized Finance (DeFi): Recreating traditional financial systems in a decentralized environment.

- **Supply Chain Management:** Providing transparent and traceable tracking of goods from origin to consumer.

⇒ **Copilot:**

Core Concept

Blockchain is a **decentralized, distributed, and immutable digital ledger** that records transactions in a chronological chain of blocks. Each block contains transaction data, a timestamp, and a cryptographic hash of the previous block, making the system tamper-evident and highly secure.

Structure & Function

- **Blocks:** Units of data containing transactions and linked cryptographically.
- **Chain:** Each block references the previous one, forming a continuous, verifiable record.
- **Immutability:** Altering past data requires recalculating all subsequent blocks, which is computationally infeasible.

Network Architecture

- Operates on a **peer-to-peer (P2P) network** where all participants (nodes) maintain synchronized copies of the ledger.
- **Consensus mechanisms** (e.g., Proof of Work, Proof of Stake) ensure agreement on the validity of new transactions across distributed nodes.
- This decentralized validation removes reliance on intermediaries and eliminates single points of failure.

Smart Contracts

Blockchain often integrates **smart contracts**, which are self-executing scripts that trigger actions automatically when predefined conditions are met. This adds programmability and automation to the ledger.

Key Features

- **Decentralization:** No central authority controls the system.
- **Transparency:** All participants can view and verify transactions.
- **Security:** Cryptographic linkage and consensus mechanisms protect against tampering.
- **Automation:** Smart contracts streamline processes without human intervention.

Applications

- **Cryptocurrency:** Digital money like Bitcoin or Ethereum.
- **Supply Chain Management:** Tracking goods with transparency and trust.
- **Decentralized Finance (DeFi):** Financial services without traditional banks.

3. Role-Based Prompt:

Prompt:

• Consider you are a Researcher in the institution and now summarize the passage in that role perspective.

Responses generated by AI Tools:

⇒ **ChatGPT:**

From a research perspective, **blockchain technology** can be described as a decentralized and distributed digital ledger system that enables secure and transparent recording of transactions across a peer-to-peer network. The architecture eliminates the dependency on a centralized authority by allowing multiple nodes to maintain synchronized copies of the ledger,

thereby enhancing system reliability, transparency, and fault tolerance. This distributed structure is a significant focus of academic and technological research because it provides a new model for secure data management in digital systems.

Technically, blockchain organizes data into sequential **blocks** that contain transaction records, timestamps, and cryptographic hashes linking each block to its predecessor. This cryptographic chaining ensures data integrity and immutability, since modifying a single block would require recalculating all subsequent blocks in the chain. From a research standpoint, this property is particularly important for studying secure data storage, distributed trust mechanisms, and tamper-resistant information systems.

The consistency of the ledger across distributed nodes is maintained through **consensus algorithms**, such as Proof of Work (PoW) and Proof of Stake (PoS). These mechanisms enable participants in the network to collectively validate transactions and agree on the current state of the ledger without relying on intermediaries. Researchers often analyze these consensus models to evaluate their efficiency, scalability, security, and energy consumption in large distributed systems.

Furthermore, blockchain platforms support **smart contracts**, which are programmable scripts that automatically execute when predefined conditions are satisfied. In research contexts, smart contracts are studied for their potential to automate complex processes, reduce operational overhead, and create trustless digital agreements.

Due to these characteristics, blockchain has become an important subject of interdisciplinary research and innovation. Its applications extend to areas such as **cryptocurrency systems, supply chain monitoring, and decentralized financial services**, where secure and transparent data management is critical. Consequently, blockchain continues to attract significant academic and industrial interest for developing next-generation distributed computing and digital transaction frameworks.

⇒ **Gemini:**

As an AI adopting the perspective of an institutional researcher, I would frame this text as a high-level overview of Distributed Ledger Technology (DLT) and its underlying cryptographic mechanisms. Here is an analytical summary suitable for a research brief or an academic abstract:

Research Brief: Architectural and Functional Overview of Blockchain Technology

- **Structural Paradigm:** Blockchain represents a shift toward decentralized, trustless architectures. By utilizing a peer-to-peer (P2P) network to maintain a synchronized distributed ledger, the technology effectively eliminates reliance on centralized authorities and mitigates single points of network failure.
- **Cryptographic Integrity:** The system achieves strict data immutability through sequential cryptographic hashing. Because each block contains a timestamp and the hash of its predecessor, the ledger is inherently tamper-evident; malicious retroactive alteration is rendered computationally infeasible.
- **Distributed Consensus Mechanism:** To ensure network-wide agreement on the ledger's state without a central arbiter, the architecture relies on consensus protocols (notably Proof of Work or Proof of Stake). These mechanisms are critical for validating state changes, ensuring fault tolerance, and preventing double-spending.
- **Programmability and Autonomous Execution:** The integration of "smart contracts" elevates the blockchain from a passive data repository to a programmable environment. These deterministic, self-executing scripts automate actions based on strictly predefined parameters, further reducing the need for trusted intermediaries.
- **Applied Implications:** The convergence of decentralization, cryptographic security, and automated logic provides a highly resilient infrastructure. Its utility is demonstrably scaling beyond foundational cryptocurrencies into complex domains such as decentralized finance (DeFi) and verifiable supply chain logistics.

⇒ **Copilot:**

As a researcher, I would frame the summary of blockchain technology in a more analytical and academic tone, emphasizing its structural principles, operational mechanisms, and broader implications:

Research-Oriented Summary of Blockchain Technology

Blockchain represents a **decentralized and immutable digital ledger system** designed to securely record and verify transactions across distributed networks. Its architecture is based on a sequential chain of blocks, each containing transaction data, timestamps, and cryptographic references to preceding blocks. This design ensures tamper resistance, as altering historical records would necessitate recalculating all subsequent blocks, which is computationally prohibitive.

The system operates through a **peer-to-peer network of nodes**, each maintaining synchronized copies of the ledger. To achieve consistency and trust without centralized oversight, blockchain employs **consensus mechanisms** such as Proof of Work (PoW) and Proof of Stake (PoS). These protocols validate new entries and safeguard against issues like double-spending, thereby eliminating reliance on intermediaries and reducing systemic vulnerabilities.

Beyond transaction recording, blockchain integrates **smart contracts**, which are self-executing scripts that enforce predefined conditions. This functionality extends the ledger's utility beyond financial applications, enabling automated processes and programmable trust.

From a research perspective, blockchain's defining attributes—**immutability, decentralization, transparency, and automation**—position it as a transformative infrastructure. Its applications span diverse domains, including **cryptocurrency systems, supply chain management, and decentralized finance**, highlighting its potential to reshape both technological and socio-economic landscapes.

This version emphasizes the **conceptual rigor and systemic implications** of blockchain, aligning with how a researcher would present it in an institutional or academic setting.

RESULT:

PROMPT TECHNIQUE	PLATFORM	ACCURACY	COHERENCE	SIMPLICITY	SPEED	UX
ZERO-SHOT PROMPTING	CHATGPT	4.2	4.5	4	5	5
	GEMINI	4.7	4.6	3	3.7	4.5
	COPILOT	3	3.9	4.2	4.1	4
FEW-SHOT PROMPTING	CHATGPT	4	4.3	4	5	5
	GEMINI	5	4.2	3	4	4.9
	COPILOT	3.3	4	4	4.3	3
CHAIN-OF- THOUGHT PROMPTING	CHATGPT	4	4.7	4	5	4
	GEMINI	4.5	4	4.2	4	5
	COPILOT	4	3.9	4	4.3	3
ROLE-BASED PROMPTING	CHATGPT	3.8	4.5	4	5	4.7
	GEMINI	4.5	4.2	4.2	4	5
	COPILOT	3.9	4	4	4	4